Computational Biology

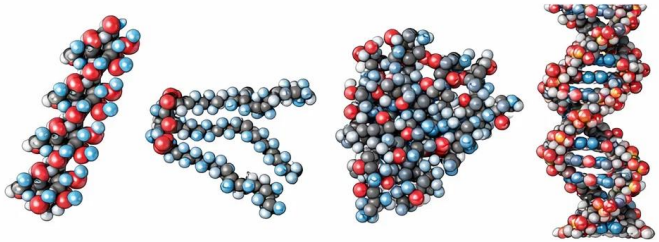
Session 2



Today's Agenda

- DNA
- Protein
- Central Dogma

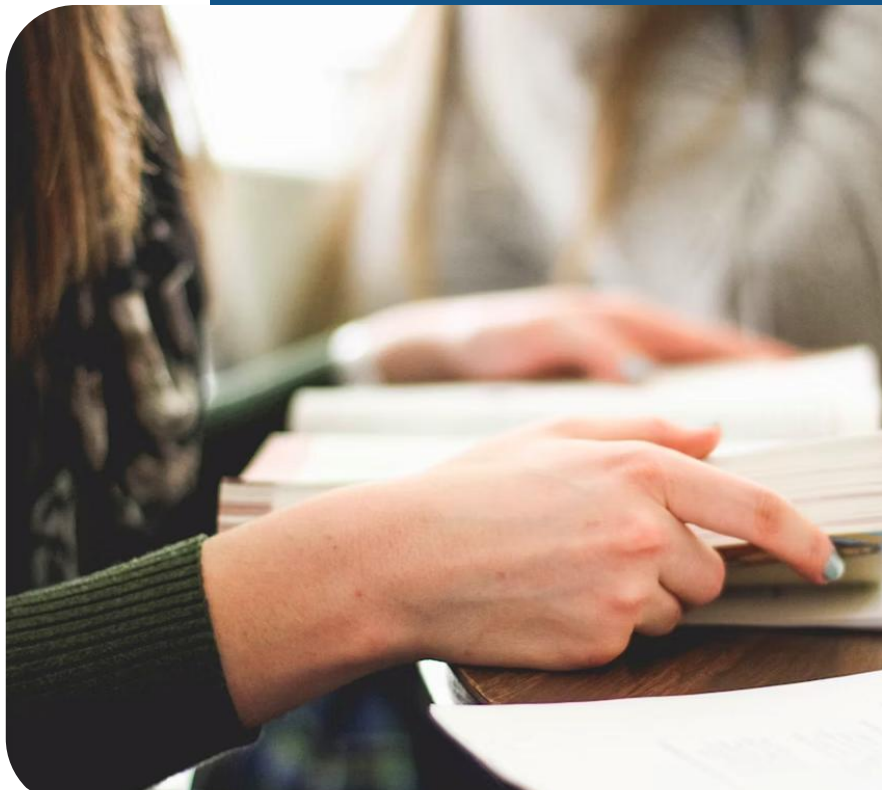
Biological Macromolecules

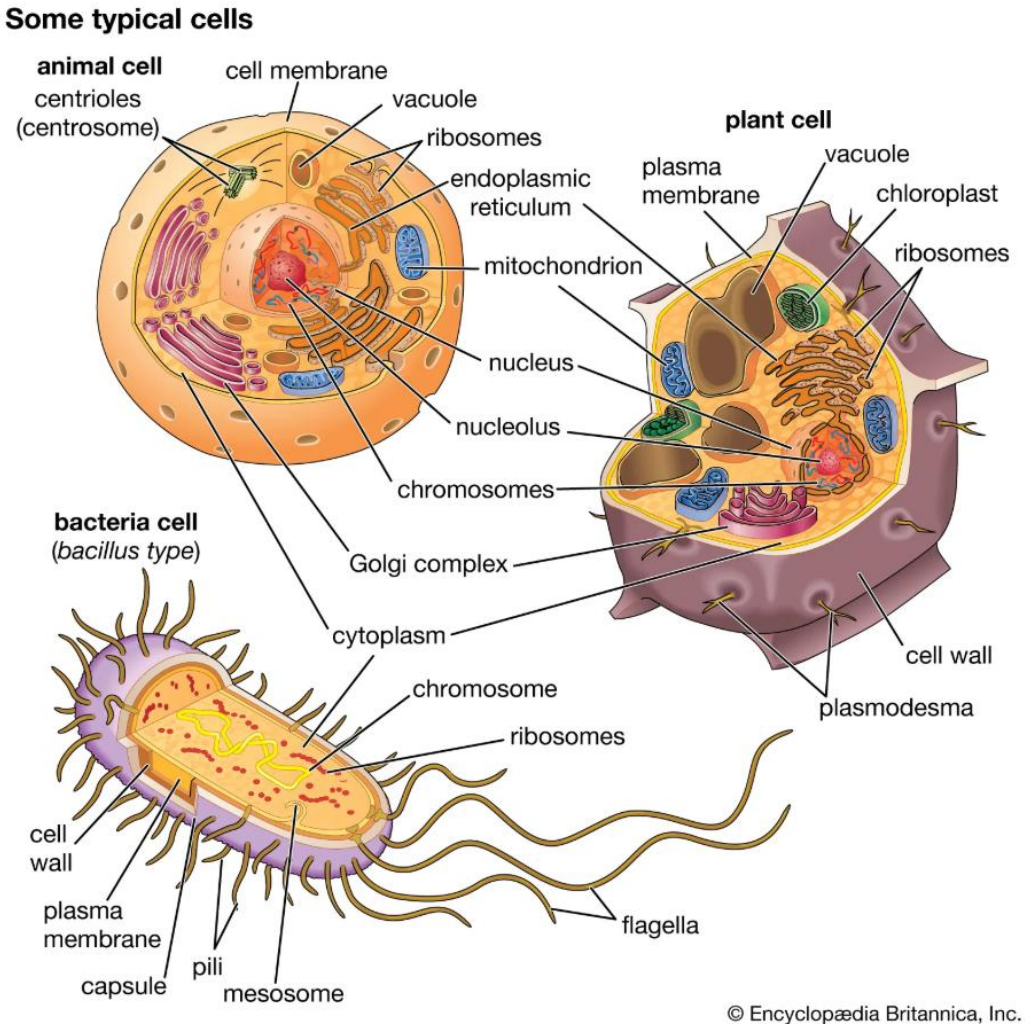


Carbohydrate (starch) | Lipid (triacylglycerol) | Protein (enzyme) | Nucleic acid (DNA)

DNA

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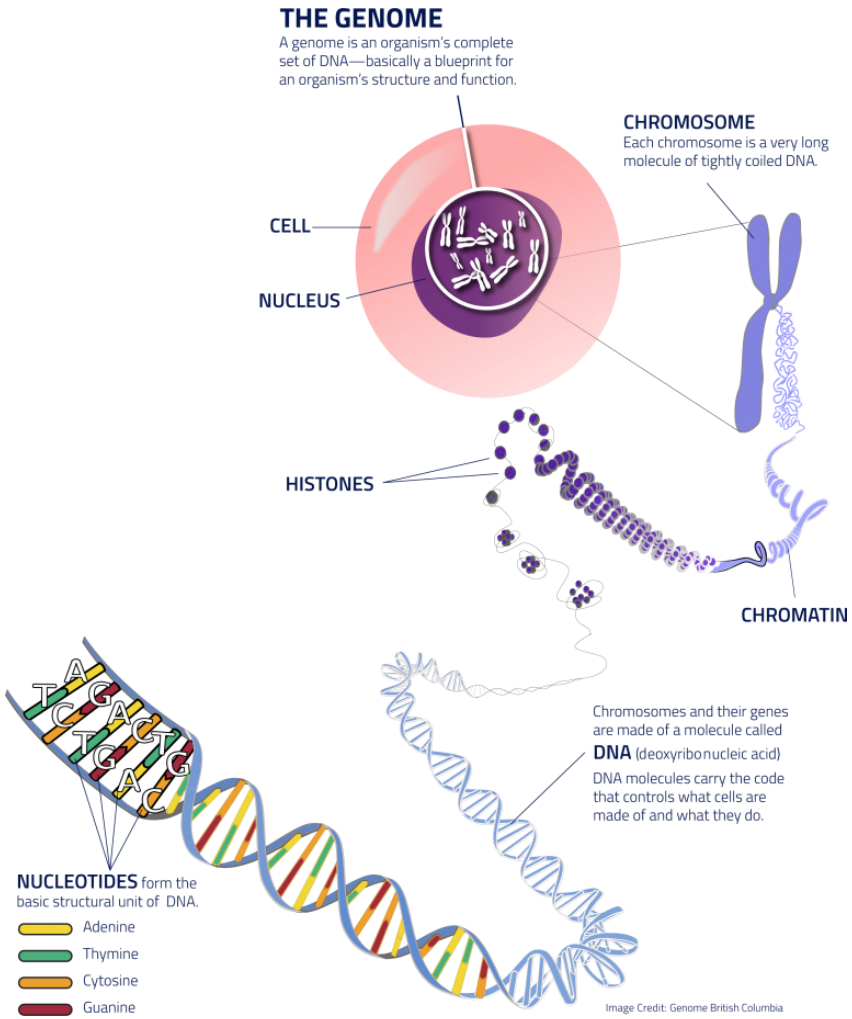




A cell is the smallest, basic, and functional structural unit of all living organisms, acting as the building block of life.

Enclosed by a membrane and filled with cytoplasm, cells contain genetic material (DNA) and are responsible for metabolism, growth, and replication.

They are broadly categorized into prokaryotes (no nucleus, e.g., bacteria) and eukaryotes (with a nucleus, e.g., plants, animals).

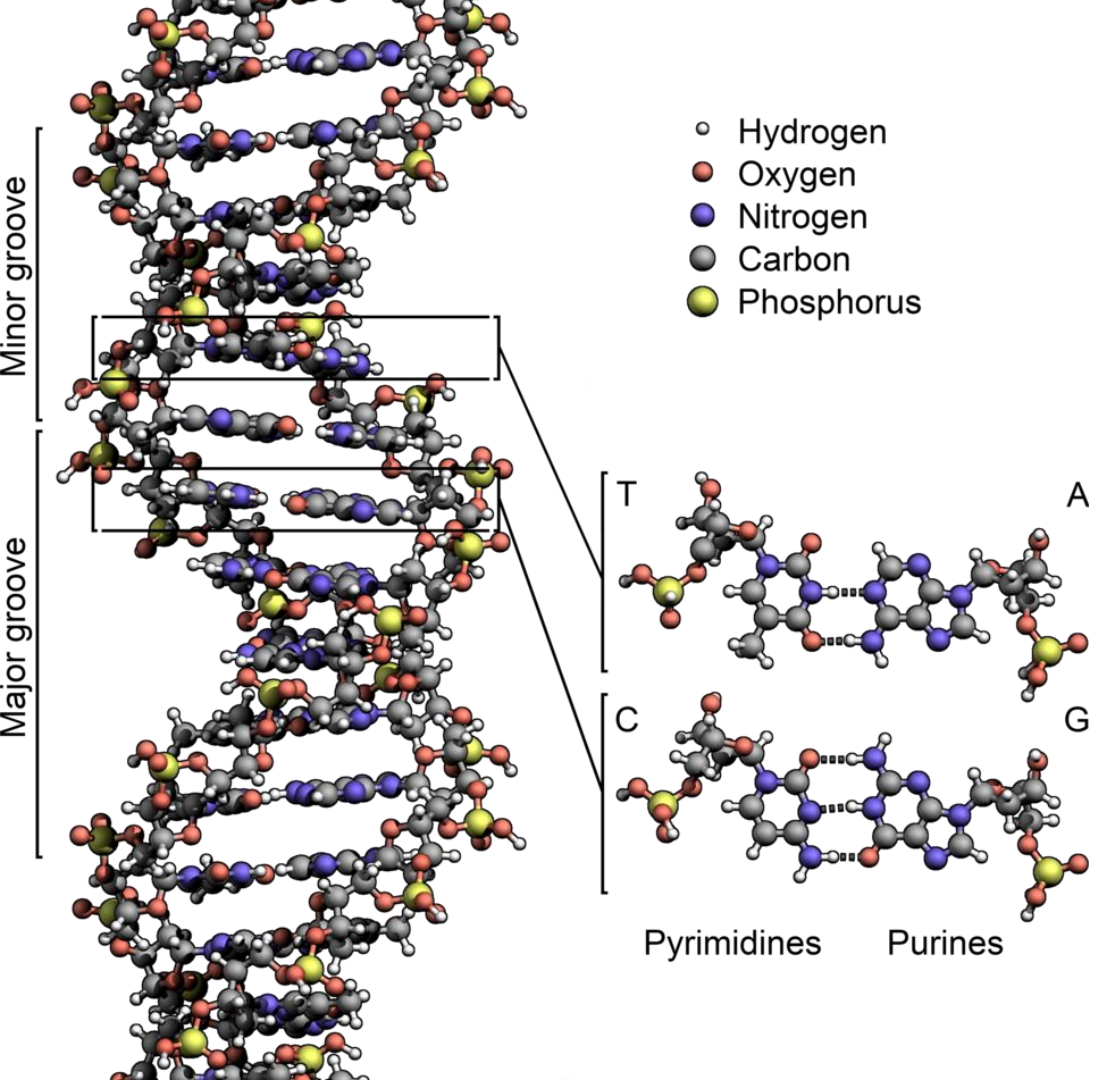


Chromosomes are organized structures inside cells that store and manage genetic material, ensuring it is properly maintained and passed on during cell division.

They are composed of long DNA molecules packaged with specialized proteins, which allow the DNA to be compacted, protected, and regulated within the cell nucleus.

DNA carries the genetic information that provides instructions for growth, development, and normal cellular function across all living organisms.

- Humans have **46 chromosomes** (23 pairs) in most cells.
- The human genome contains about **3.2 billion DNA base pairs**.
- Humans have roughly **20,000–21,000 genes**, making up only **~1–2%** of all DNA.
- If stretched end to end, DNA from a single human cell is about **2 meters long**.
- Almost every cell in the body contains the **same DNA**, but different genes are turned on or off.



DNA carries the genetic information that provides instructions for growth, development, and normal cellular function across all living organisms.

Its sequence contains instructions for making proteins and functional RNAs that carry out most cellular processes.

By controlling which molecules are produced and when, DNA helps determine cell behavior, development, and inherited traits.

THE STRUCTURE OF DNA

THIS PART

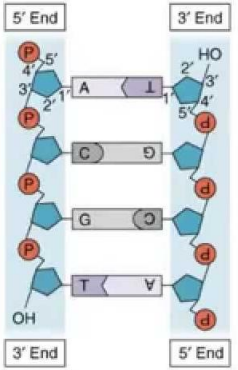
'DOUBLE HELIX'

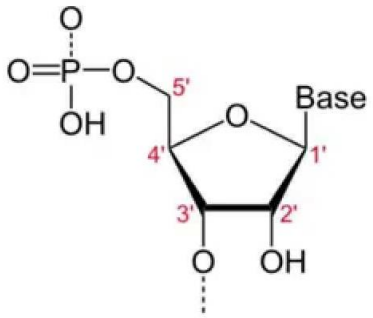
BASES ALL STICK OUT

THESE ARE WHAT HOLD THE TWO STRANDS TOGETHER

PAIR UP

ONLY COMPLEMENTARY BASES CAN PAIR:





5' AND 3' ENDS OF NUCLEIC ACIDS

Also read the orientation of a DNA strand (3'-5')

THE STRUCTURE OF DNA

